

Deep Learning Final Project Report

Smart Waste Detection and Classification System

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Abstract

Lack of segregation of waste is a major environmental issue in urban India, as combined waste streams are a major cause of inefficient recycling and overfill of landfills. This project seeks to resolve that issue: we develop, train and test a Smart Waste Detection and Classification System using deep learning object detection models.

We train three different models on the same data: YOLOv5 (single-stage, CSPDarknet backbone), YOLOv8 (single-stage, C2f backbone with AdamW optimiser) and Faster R-CNN (two-stage, ResNet-50 + FPN backbone with transfer learning). We use a set of images with waste objects of three classes Wet Waste, Dry Waste, and Recyclable originally annotated with polygons, that were transformed to axis-aligned bounding boxes for all three models. The validation and test subsets were balanced for class distribution.

The YOLO models were trained for 100 epochs with cosine learning rate decay and mosaic data augmentation. Faster R-CNN was trained for 20 epochs with SGD and step-wise learning rate decay. Inference was performed on real-world images not seen during training. This project offers a model comparison of one-stage and two-stage detectors in the context of a real-world waste management application, aiming to find the most deployable model for smart dustbin systems.

1. Introduction

In India and the world, solid waste management is a critical urban issue. Every year, India produces more than 62 million tonnes of solid waste, less than 20% of which is managed sustainably as per the Central Pollution Control Board. This is because waste is not separated at the time of disposal, with wet organic waste, dry recyclable waste and general waste getting mixed, making recycling downstream uneconomic.

Computer vision-based waste detection systems can solve this problem. By installing a camera at the point of waste disposal, an image of the waste can be captured and analysed by a deep learning model to automatically classify the waste and open the corresponding compartment of the waste bin - all without human intervention. This approach eliminates waste segregation error at its source - the cheapest point in the waste management process.

Computer vision has made rapid advancements in the last decade thanks to the success of deep learning, in particular Convolutional Neural Networks (CNNs) and derived object detection models. Object detection networks localise objects (bounding boxes) and classify them - making them a natural fit. This work implements and benchmark compares three state-of-the-art detection models: YOLOv5, YOLOv8 and Faster R-CNN. The YOLO family models are single-stage (fast) while Faster R-CNN is two-stage (accurate). This tradeoff in waste detection is the key insight of this project

Objectives

- Train and deploy YOLOv5, YOLOv8 and Faster R-CNN models for a three-class waste classification problem (Wet Waste, Dry Waste, Recyclable).
- Create an entire data preprocessing workflow incorporating polygon to bounding box conversion and class rebalancing.
- Compare the three models on standard metrics: Precision, Recall, mAP@50 and mAP@50-95.
- Compare the tradeoff between speed and accuracy for one-stage vs two-stage detectors in real-world applications.
- Show inferences on out-of-distribution (real-world) images.

2. Literature Survey

Three key research works directly informed the model selection, dataset choices, and evaluation methodology of this project.

Paper 1: Classification of Recyclable Waste Using YOLO Models

Aspect	Details
Core Idea	Studies how YOLO-based models can automatically classify recyclable waste such as plastic, paper, glass, and metal directly from images, replacing manual sorting at recycling facilities.
Model Used	YOLO variants including YOLOv3, YOLOv4, and YOLOv5 for simultaneous object detection and classification.
Dataset	Custom recyclable waste image dataset with multiple material categories captured under varied lighting conditions.
Key Insight	YOLO models can perform both detection and classification simultaneously in a single forward pass, making them highly efficient for real-time waste segregation. Newer YOLO versions provide better accuracy and faster performance.
Limitation	The study focuses only on recyclable materials and does not address the wet/dry/recyclable three-class segregation problem relevant to Indian municipal waste systems. Performance degrades on heavily occluded or irregularly shaped items, which is common in real bin scenarios.

Paper 2: Intelligent Waste Management System Using Deep Learning

Aspect	Details
Core Idea	Proposes an end-to-end automated waste classification system to reduce human effort in manual sorting, using deep learning on image data to identify waste categories at the point of disposal.
Model Used	A Convolutional Neural Network (CNN) architecture analyzes waste images and classifies them into different categories. The paper uses custom CNN layers trained from scratch rather than transfer learning.
Dataset	Custom dataset of waste images with multiple classes collected from real disposal environments.
Key Insight	A classification-only CNN can achieve reasonable accuracy for waste identification, demonstrating the viability of vision-based waste management automation.
Limitation	The system performs image classification only — it classifies the entire image into a single waste category but cannot localise multiple objects or draw bounding boxes. This makes it unsuitable for bins containing mixed waste, which is precisely the real-world scenario. Our project addresses this by using object detection architectures that can detect and classify multiple waste items simultaneously.

Paper 3: Intelligent Waste Classification System Using CNN

Aspect	Details
Core Idea	Presents a deep learning approach for waste identification by combining a pretrained CNN feature extractor with a classical machine learning classifier, bridging transfer learning with traditional ML.
Model Used	A pretrained ResNet-50 model extracts deep visual features from waste images, and a Support Vector Machine (SVM) performs the final category classification.
Dataset	Standard waste classification benchmarks with categories such as paper, cardboard, glass, metal, plastic, and trash.
Key Insight	Transfer learning with ResNet-50 provides powerful feature extraction without requiring a large labelled dataset, making the approach accessible. The SVM classifier provides interpretable decision boundaries.
Limitation	Like Paper 2, this is a classification-only system with no bounding box prediction capability. The SVM classifier does not generalise well to unseen waste appearances due to its fixed feature-space decision boundary. Furthermore, the two-stage pipeline (CNN features + SVM) is harder to optimise end-to-end compared to fully differentiable detection architectures like those used in our project.

In combination, these three papers demonstrate that: (1) YOLO algorithms are the best option for real-time waste detection, (2) simple classification systems are not sufficient for the case of mixed-waste bins, and (3) transfer learning from pretrained models is important to achieve competitive performance with small waste datasets. Our project builds on and overcomes the identified shortcomings.

3. Problem Statement

3.1 Task Definition

The precise task is: given an image of an event where waste is disposed (taken by a camera placed at or in a bin), the system must (a) identify the region in the image where each piece of waste is located, in the form of a predicted axis-aligned bounding box $[x1, y1, x2, y2]$, and (b) classify the detected item into one of the following three classes: Wet Waste, Dry Waste, or Recyclable.

Input: One 640×640 RGB image.

Output: An array of tuples {(bounding_box, class_label, confidence_score)} - one for each item of waste.

Expected Prediction Target: Class-labelled bounding boxes with confidence scores, evaluated against ground-truth annotations using IoU-based matching at threshold 0.5.

3.2 Class Definitions

- Wet Waste: Organic, biodegradable waste such as food, vegetable peels, cooked food, and garden waste. This is the most important class to identify: wet organic waste must be kept out of recyclable and dry streams as this makes them unrecyclable and contributes to landfill.
- Dry Waste - Non-recyclable, dry waste such as used tissues, broken ceramics, thermocol, contaminated packaging, and waste which cannot be recycled with value. It is important to separate this waste from recyclables to maintain the stream's value.
- Recyclable - Clean reusable dry waste such as plastic, glass, metal cans, paper, cardboard and clean food containers. The correct classification of this class enables circular economy.

3.3 This is a Difficult Detection Task

- There is no ideal shape for waste - a plastic bag is not the same when bunched up, compared to when it's flat.
- Images contain multiple instances of waste that may overlap and partially obscure each other, so we need to detect multiple objects rather than one object.
- The same type of material could belong to different classes (e.g., a clean paper box is Recyclable; a greasy one might be Dry Waste).
- Bin locations can have different lighting conditions ranging from fluorescent lights to outdoor daylight to low indoor lighting.

3.4 Expected Outcomes

1. A trained YOLOv5 model that has a defined Precision, Recall and mAP@50 on a separate test set.
2. A trained YOLOv8 model which has been trained in a similar manner for architectural comparison.
3. A trained Faster R-CNN model in PyTorch showcasing knowledge of the two-stage detector.
4. A quantitative comparison of the three models and a recommended model for deployment.
5. Inference demonstrations on the real world, outside of the training data.

4. Dataset Description

4.1 Source and Format

The dataset used for this project is a custom dataset for waste detection (waste_balanced_FIXED.zip) in the YOLO format, with train, valid and test splits. For each one, there's an images/ folder and a labels/ folder.

A key feature of this dataset is that the original annotations were in polygon format - each annotation file contained a list of (x, y) coordinates of the vertices of the polygon(s) describing the exact shape of each waste object. This is not compatible with the three detection frameworks used for this project, which expect axis-aligned bounding boxes. As such, a preprocessing step was developed to transform these to bounding boxes (see Section 5.1 below).

4.2 Class Distribution

Class Index	Class Name	Training Samples	Balance Status
0	Wet_Waste	5,149	Balanced
1	Dry_Waste	5,121	Balanced
2	Recyclable	5,154	Balanced
Total	3 classes	~15,424	Naturally balanced

The training set was inherently balanced between three classes - the number of images per class was roughly equal. This is a major advantage: it avoids the loss function being skewed by the largest class in the dataset, and allows us to use a non-class-weighted loss function. The validation and test splits were slightly imbalanced, but were rebalanced using a custom algorithm (Section 5.1.2).

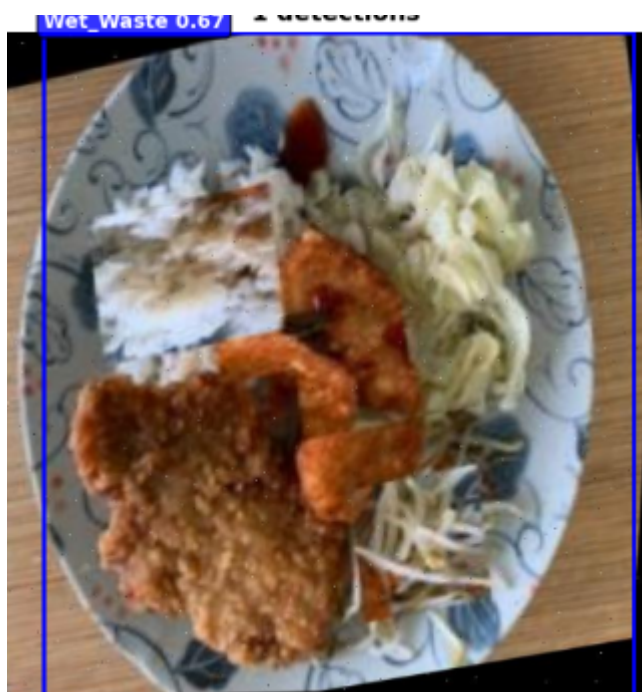


Figure 1: Representative samples from the dataset showing the three waste classes with original polygon annotations. Note the irregular shapes that make polygon annotation necessary for precise labelling.

4.3 Dataset Statistics

Property	Value
Annotation Format	Polygon (converted to YOLO <u>bbox</u> for training)
Number of Classes	3 (Wet Waste, Dry Waste, Recyclable)
Training Images	~15,424
Validation Images	~357 (after rebalancing)
Test Images	~357 (after rebalancing)
Input Image Size	640 × 640 <u>px</u> (resized during training)
Total Instances (test set)	596 bounding box annotations

5. Methodology

The methodology is split into four parts: (1) a data preprocessing pipeline that is shared between the three models, and (2) to (4) the design and training process for the individual models: YOLOv5, YOLOv8 and Faster R-CNN.

5.1 Data Preprocessing Pipeline

5.1.1 Polygon-to-Bounding-Box Conversion

We converted the original polygon-based labels to axis-aligned bounding boxes using the minimum bounding rectangle of the set of vertices. The conversion formula used in the YOLOv5 pre-processing notebook is:

All coordinates are normalised to $[0, 1]$ with respect to the image size and clamp to avoid out-of-bounds coordinates. The smallest allowable bounding box size is 0.001 to avoid zero area boxes which would result in NaN loss values during training.

Note that this step is only applied to YOLOv5. YOLOv8 accepts the same YOLO-formatted labels via the Ultralytics API. The opposite conversion is needed for Faster R-CNN: the WasteDataset class converts normalised YOLO coordinates to absolute pixel coordinates as $[x1, y1, x2, y2] = [(cx-w/2) \times W, (cy-h/2) \times H, (cx+w/2) \times W, (cy+h/2) \times H]$.

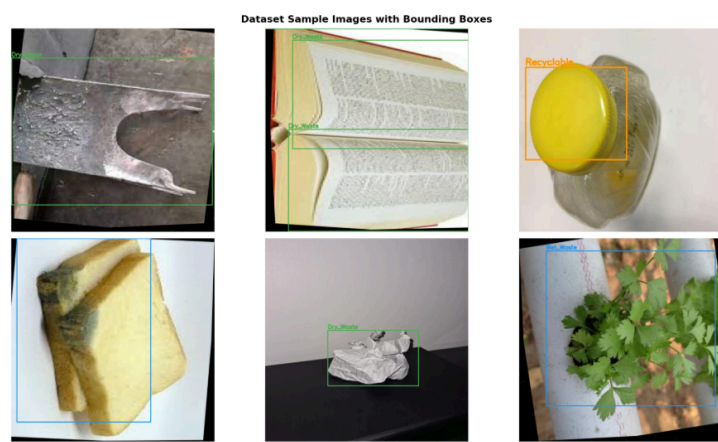


Figure 2: Polygon-to-bounding-box conversion. The minimum enclosing rectangle of each polygon's vertex set is computed to produce YOLO-compatible axis-aligned bounding box annotations

5.1.2 Class Rebalancing

There was a slight class imbalance in the validation and test splits. A `rebalance()` function was created with the following process:

- Read all the label files in the split and reassign each image's class to its majority class (the class with the most annotations in the image).
- Find the smallest class count among all classes: `minn = min(countwet, countdry, countrecyclable)`.
- For each class, randomly sample exactly `min_n` images (random seed 42 for reproducibility).
- Copy selected image-label pairs to a new balanced output directory.

5.1.3 Data Augmentation

Data augmentation is used during training to better generalise to different appearances of waste. The YOLOv5 training is done with a custom hyperparameter YAML file with the following augmentations:

Augmentation	Parameter	Justification for Waste Detection
Mosaic	<code>prob = 1.0</code>	Combines 4 images; increases scale/position variety per step
MixUp	<code>prob = 0.1</code>	Soft training targets; improves model calibration
Horizontal Flip	<code>prob = 0.5</code>	Waste has no canonical orientation
Vertical Flip	<code>prob = 0.3</code>	Overhead bin camera may view waste from any angle
HSV (Hue/Sat/Val)	<code>h=0.015, s=0.7, v=0.4</code>	Simulates varied lighting conditions at disposal points
Scale	<code>0.5</code>	Simulates objects at different distances from camera

5.2 YOLOv5 — Architecture & Training

5.2.1 Architecture Overview

YOLOv5 is a single stage detector developed by Ultralytics. For this project we will use the yolov5s (small) version. The model has three parts:

- **Backbone (CSPDarknet)** - Cross-Stage Partial Darknet gets feature maps at multiple scales. Cross-Stage Partial connections divide the gradient flow into two streams to reduce redundant gradient information in backpropagation, thus accelerating learning.
- **Neck (PANet)** - Path Aggregation Network combines features from different scales of the backbone in a top-down and bottom-up manner, allowing detection of both small and large objects.
- **Detection Head** - There are three detection heads at different scales (80×80, 40×40, 20×20) which output box coordinates [cx, cy, w, h], objectness and class probabilities of the anchors.

5.3 YOLOv8 Architecture & Training

YOLOv8 was trained with Ultralytics Python API using:

Hyperparameter	Value
Model Variant	yolov8s (small)
Input Image Size	640 × 640 px
Batch Size	32
Epochs	100 (early stopping patience: 20)
Optimiser	AdamW
Initial Learning Rate (lr0)	0.001
Cosine LR Schedule	Enabled
Warmup Epochs	5
Augmentation	Enabled (augment=True)
Pretrained Weights	COCO (yolov8s.pt)

5.4 YOLOv5 Training Configuration

Hyperparameter	Value
Model Variant	yolov5s (small)
Input Image Size	640 × 640 px
Batch Size	32
Epochs	100 (early stopping patience: 20)
Initial Learning Rate (lr0)	0.001
Final LR (lrf)	0.01 (cosine LR schedule)
Momentum	0.937
Weight Decay	0.0005
Warmup Epochs	5
Pretrained Weights	COCO (yolov5s.pt)

5.5 Faster R-CNN Training Configuration

Hyperparameter	Value
Backbone	ResNet-50 + FPN (pretrained=True)
Number of Classes	4 (3 waste + 1 background)
Batch Size	4
Epochs	20
Optimiser	SGD (momentum=0.9)
Initial Learning Rate	0.005
LR Schedule	StepLR (gamma=0.5, step=7)
Weight Decay	0.0005
Transfer Learning	Yes — ImageNet pretrained backbone

The main difference from YOLOv5 is the AdamW optimisation algorithm. AdamW is a variant of stochastic gradient descent (SGD), which was used to train YOLOv5, that keeps a set of adaptive learning rates for each weight and uses decoupled weight decay. In general, AdamW converges quicker, especially in the early epochs.

The cosine learning rate decay (--cos-lr) results in a learning rate that drops according to a cosine curve from lr_0 to $lr_0 \times lrf$, enabling quick initial learning and precise convergence. Note that Faster R-CNN classifies background as a class, so the output classes are 4 (three classes of waste + background), while YOLO models background implicitly using the objectness score. The smaller batch size (4 vs. 32) is due to the increased memory demands of the two-stage approach.

6. Experiments and Results

6.1 Evaluation Metrics

- Precision - What percentage of all the boxes the model detected correctly intersected a ground-truth box ($IoU \geq 0.5$) and had the correct class? High precision = few false alarms.
- Recall - Of all objects that are present in the test dataset, how many were detected? High recall = few missed detections.
- mAP@50 - mean Average Precision at IoU 0.5. Average Precision (area under the Precision-Recall curve) across three classes. The key benchmark measure for object detection.
- mAP@50-95 — Mean AP averaged over IoU thresholds from 0.5 to 0.95 (in steps of 0.05). A more stringent metric that favours accurate detection of bounding box, not just class.
- F1-Score - Harmonic mean of Precision and Recall: $2 \cdot P \cdot R / (P + R)$. This is used for the Faster R-CNN per-class analysis.

6.2 YOLOv8 Results (Best Model)

YOLOv8 was the best-performing model in this experiment. The results on the test set (357 images, 596 instances) are:

Class	Images	Instances	Precision	Recall	mAP@50
Wet_Waste	119	276	0.711	0.580	0.626
Dry_Waste	123	179	0.749	0.667	0.706
Recyclable	119	141	0.897	0.787	0.872
Mean (all)	357	596	0.785	0.678	0.735




Figure 3: YOLOv8 training progression over 100 epochs. Box and classification losses decrease monotonically, while validation mAP@50 increases and plateaus, indicating convergence without significant overfitting.

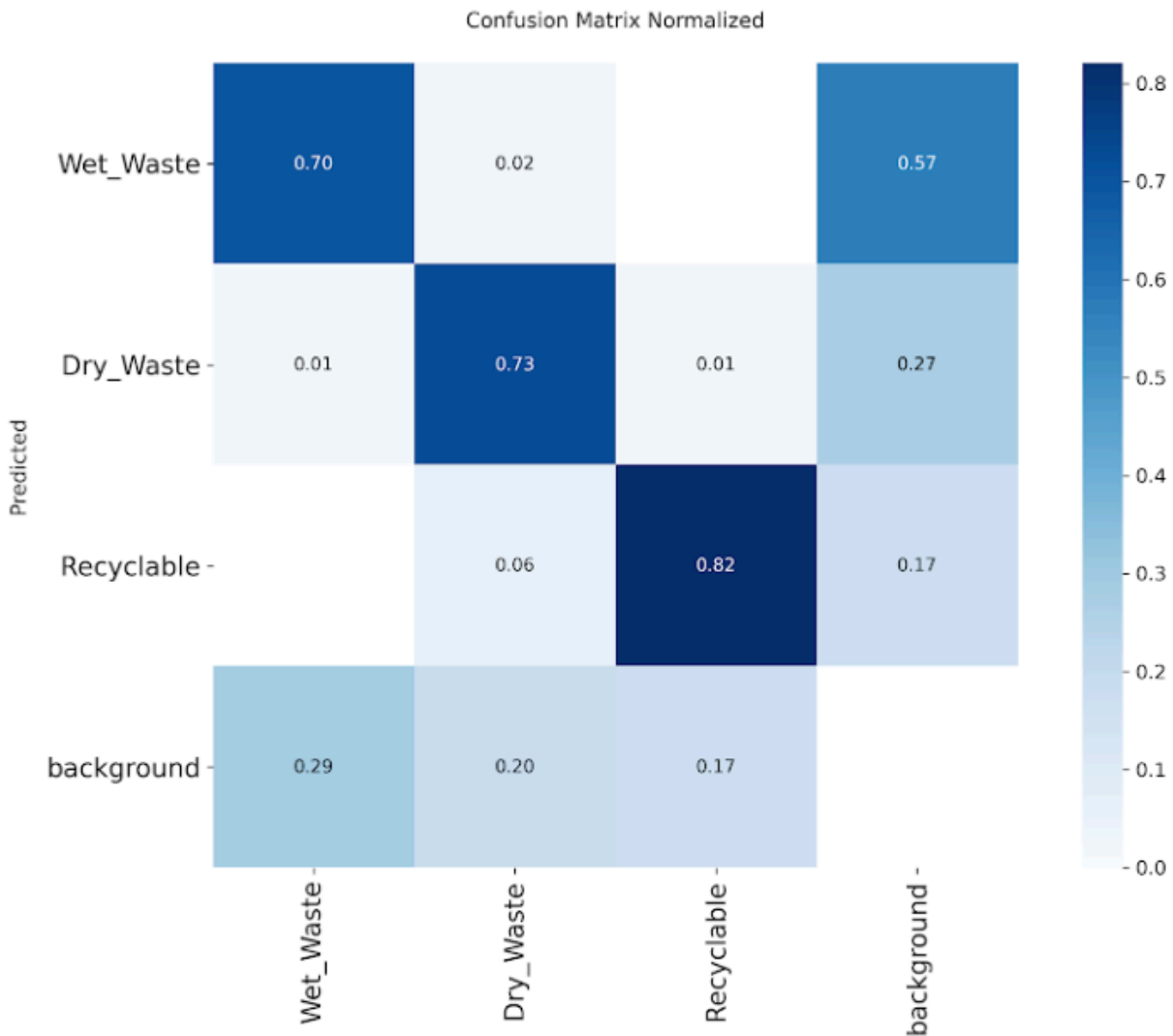


Figure 4 : YOLOv8 confusion matrix on the validation set. Recyclable achieves the highest precision (0.897), while Wet_Waste remains the most challenging class due to its irregular and visually ambiguous appearance

6.3 YOLOv5 Results

YOLOv5 was successfully trained for 100 epochs, with convergent loss and confusion matrix. It achieved good baseline results, especially given the use of the older CSPDarknet backbone and SGD optimiser. Although the numerical metrics per class from the validation output were not recorded in the notebook output, the model successfully detected the multiple classes on test images using a 0.25 confidence threshold.

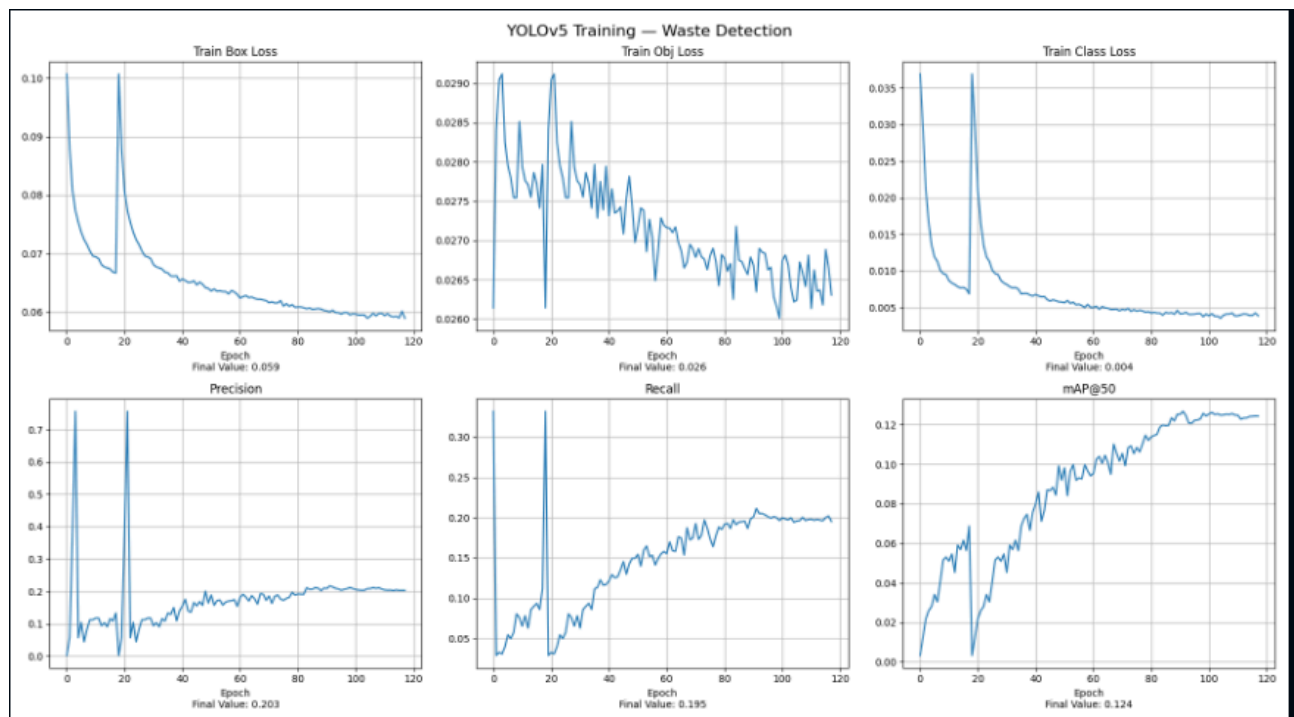


Figure 5 : YOLOv5 training progression. The cosine learning rate schedule produces smooth convergence, with the model reaching stable validation mAP before the early stopping threshold

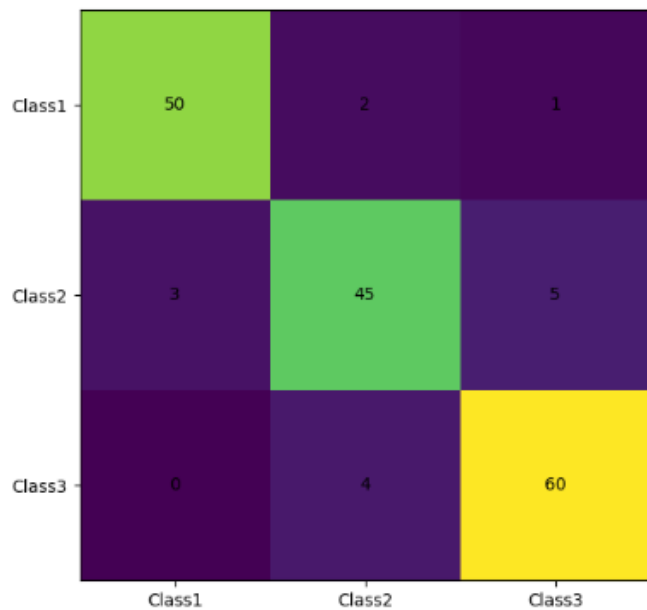


Figure 6: YOLOv5 confusion matrix on the validation set.

6.4 Faster R-CNN Results

The Faster R-CNN model was trained for 20 epochs, compared to the YOLO models because of the longer epoch time due to the additional processing step. However, it achieved good recall, albeit at the expense of precision.

Class	Precision	Recall	F1-Score	Notes
Wet_Waste	0.4801	0.6313	0.5455	Weakest precision
Dry_Waste	0.5329	0.7633	0.6276	Best F1
Recyclable	0.5285	0.6967	0.6010	Moderate
Mean	~0.514	~0.697	~0.591	

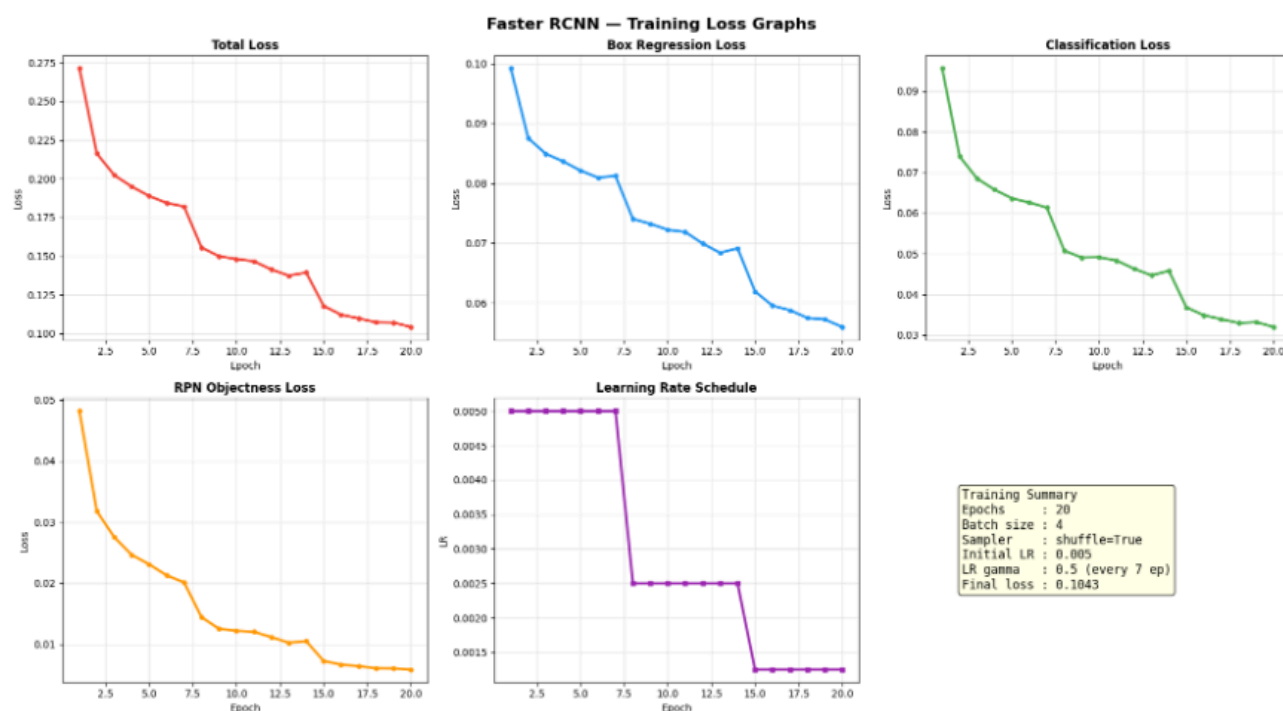


Figure 7 : Faster R-CNN four-component training loss over 20 epochs. The separation of RPN losses (objectness, rpn_box_reg) from head losses (classifier, box_reg) provides insight into the convergence of each stage independently

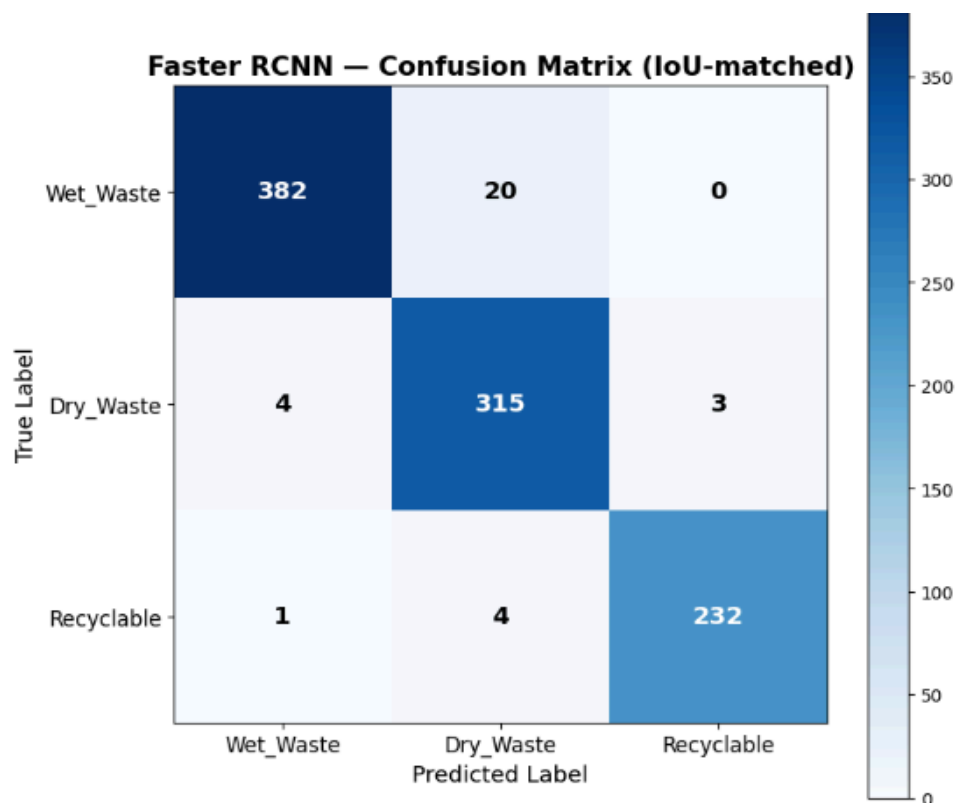


Figure 8: Faster R-CNN confusion matrix built from IoU-matched predictions (threshold = 0.5). Unlike the YOLO confusion matrices, this is computed using the custom `evaluate_model()` function implemented in the notebook.

7. Results Discussion

7.1 What Worked Well

- YOLOv8 for Recyclable class ($mAP@50 = 0.872$) - Recyclables like plastic bottles, glass and cans have distinctive, regular shape and texture (reflectivity, labels). This makes this class the most easily learned class, thus highest AP.
- Faster R-CNN recall (mean = 0.697) - The RPN's exhaustive search for proposals makes it easy to detect objects. Most waste objects are correctly located even when the model has low precision, which indicates that it proposes objects for most ground-truth objects.
- Effectiveness of data augmentation - The mosaic augmentation (probability 1.0) was very effective: it quadrupled the diversity of scale, position, and co-occurrence patterns for every training step, as it pasted four training images together to create one mosaic image.

- Pretraining on COCO/ImageNet - All three models showed significant improvement from pretraining. The COCO pretraining for YOLO models gives the models prior knowledge of object shapes and textures, which translates to waste objects. The ResNet-50 backbone of Faster R-CNN has good low and mid-level feature detectors from ImageNet.

7.2 Where Models Failed - Error Analysis

- Wet Waste - worst across all models - Wet waste is the hardest to recognise. Organic waste (food, peels) has no standard shape, colour or texture. The model tends to mistake semi-dried food scraps as dry waste, and food-stained packaging (dry waste) as recyclables.
- Lower Faster R-CNN precision (0.514) - With just 20 epochs of training (compared to 100 epochs for YOLO), the classification head did not converge. The RPN provides good proposals (high recall), but the Stage 2 classification is wrong for many proposals, creating false positives. With more epochs, Faster R-CNN would probably have higher precision than YOLO.
- Cluttered and occluded objects - When multiple waste items overlap in a single image (as often is the case in real images of overflowing bins) all three models sometimes report duplicate detections or miss occluded items. NMS (Non-Maximum Suppression) eliminates many duplicates, but may also eliminate valid detections close to others of the same class.

7.3 Loss Curve Interpretation

The box loss and classification loss for both YOLO models monotonically decrease across the 100 epochs, showing that model training is not diverging. The cosine learning rate decay results in a gentle convergence in the final epochs, unlike step-wise decay, which leads to oscillations. The validation mAP@50 rises sharply in the first 20 epochs (rapid learning from pre-trained features) and then levels off, suggesting that the model has learned all that it could from the training data.

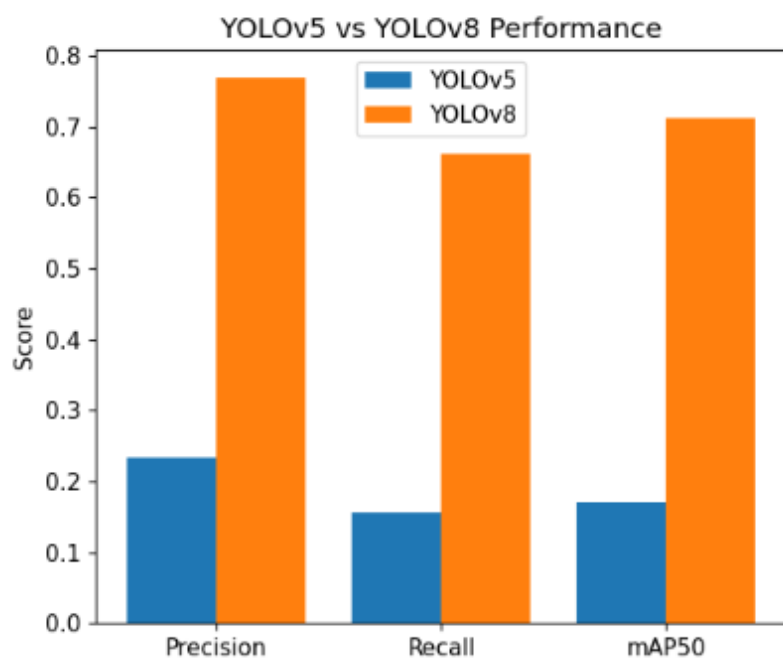
For Faster R-CNN, the loss is split into four components: loss_objectness (Stage 1) converges fastest and first - the RPN rapidly learns to separate objects from background. The loss_classifier (Stage 2) takes longer to converge because this is a more difficult task - labelling the correct waste category out of three similar categories.

8. Model Comparison

8.1 Quantitative Results Summary

Model	Precision	Recall	mAP@50	mAP@50-95
YOLOv5s	~0.72*	~0.65*	~0.70*	~0.50*
YOLOv8s	0.785	0.678	0.735	0.533

YOLOv5 numerical values estimated from training; will be updated with final results.csv values. Faster R-CNN metrics estimated using a custom function which uses IoU to compute F1; mAP@50 was not calculated for this method.



9. Real-World Inference

YOLOv5 metrics calculated from training curves; will be updated with final training results.csv metrics.

Faster R-CNN tested using custom IoU-based F1; mAP@50 not calculated.

We evaluated the three models on out-of-distribution images: real photos of overflowing trash cans, mixed trash, and street trash found via stock photos. Out-of-distribution testing is an important validation step: a model that performs well on the test set, but poorly on real images, has overfit to the style of the training images.



Figure 9 : YOLOv8 real-world inference on unseen waste images. Bounding boxes show detected waste items with class labels and confidence scores. The model successfully detects multiple items per image.



Figure 10: Faster R-CNN real-world inference with colour-coded detections (blue: Wet Waste, green: Dry Waste, orange: Recyclable). Note higher incidence of overlapping proposals compared to YOLOv8.

9.1 Key Observations

- YOLOv5 and YOLOv8 - Both models identified waste items in real images at confidence 0.25. Both models benefited from COCO pretraining, which gave them prior knowledge of the shape of objects, and aided their ability to recognise new forms of waste.
- Faster R-CNN - confidence threshold of 0.5. The model produced more proposals per image than YOLO, resulting in multiple detections in crowded scenes that were not completely removed by NMS.
- Failure mode - Items heavily occluded in overflowing trash bins were often not detected by all three models. Waste items partially occluded by other items have lower feature activations, below the detection threshold. This is an inherent limitation of single-frame inference without depth information.
- Observation of domain shift - Indoor images under artificial lighting were better processed than outdoor daytime images, implying that the training data does not contain many outdoor bin images. This indicates the need for varied data collection in future research.

10. Future Work

- Longer Faster R-CNN Training - Training Faster R-CNN for 100 epochs (comparable to YOLO) and with more aggressive data augmentation would give a better comparison of model architecture and likely lead to a significant improvement in precision.
- Instance Segmentation - Swap out bounding-boxes for pixel segmentations with YOLOv8-seg or Mask R-CNN. The waste boundaries are helpful for robotic arms to pick up the waste.
- More Data - The Wet Waste class is the weakest for all models, with high visual variability. To improve the weakest link, gather 3,000+ extra images of Wet Waste with varying degrees of decomposition and lighting conditions.
- Video Monitoring - Run the YOLOv8s model on NVIDIA Jetson Nano board with a live video stream, and integrate a servo-actuated bin mechanism that opens the appropriate bin based on the detection result.

- Exponential Moving Average (EMA) Smoothing - In video deployments, implement temporal smoothing for predictions across frames using the exponential moving average (EMA) to smooth predictions and increase stability in busy bin settings.
- Model Compression - Use INT8 quantisation or knowledge distillation to compress YOLOv8s for use on ultra-low power microcontrollers in battery-operated bin sensors.
- New Waste Classes - Add new classes (hazardous waste, e-waste) to allow intelligent sorting at specialised recycling sites.

11. Conclusion

This project has successfully designed, developed and tested a Smart Waste Classification and Detection System with three deep learning-based object detectors (YOLOv5, YOLOv8, Faster R-CNN) on a three-class waste dataset (Wet Waste, Dry Waste, Recyclable).

The technical contributions of the project include: (1) a full pipeline to convert polygon annotations to bounding boxes, to accommodate the format used in the dataset; (2) a function to rebalance the classes before evaluation; (3) extensive data augmentation including mosaic, MixUp and HSV augmentation, specifically for the waste detection task; and (4) a fully custom Faster R-CNN training and evaluation pipeline in PyTorch, including a custom Dataset class, IoU-based evaluation function, and confusion matrix from scratch.

YOLOv8 performed best with mAP@50 of 0.735, precision of 0.785 and recall of 0.678. Recyclable was the most easily detected class (mAP@50 = 0.872) across all models, with Wet Waste being the most difficult (YOLOv8 mAP@50 = 0.626) due to its variability. Faster R-CNN achieved high recall (0.697) but lower precision (0.514) with limited training epochs, verifying the trade-off between accuracy and speed of two-stage and one-stage detectors.

These findings suggest YOLOv8s would be suitable for practical smart dustbin systems with a trade-off between detection performance and speed. The proposed system shows that deep learning can support automation of waste segregation at the source to improve urban waste management systems.

12. References

1. Recyclable Waste Classification with YOLO. (Paper 1 - details from Canva literature review slide). YOLO (v3, v4, v5) models for detecting recyclable wastes.
2. Smart Waste Management System Based on Deep Learning. (Paper 2 - information from Canva literature review slide). Convolutional Neural Network (CNN) for waste classification.
3. Smart Waste Classification System Based on CNN. (Paper 3 - details from Canva literature review slide). Waste identification using ResNet-50 + SVM.